Does Educational Technology Narrow or Widen Achievement Gaps?
Cross-National Evidence and Heterogeneous Effects in India

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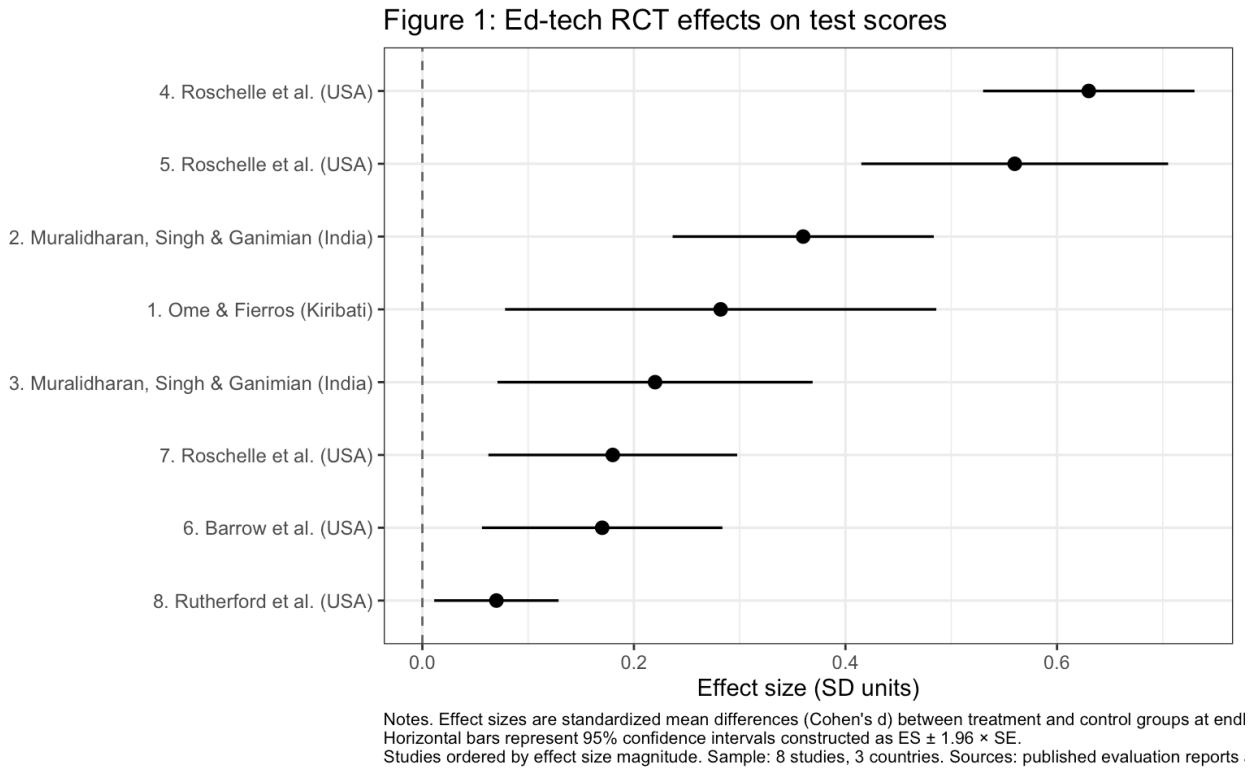
Abstract

This paper examines whether educational technology interventions narrow or widen achievement gaps, combining two complementary analyses. Part I uses a mini meta-analysis of eight randomized controlled trials across three countries to assess whether ed-tech effects are larger in lower-infrastructure settings. Using electricity access — a variable directly on the causal pathway for program implementation — I find a consistent negative gradient: lower infrastructure is associated with larger learning gains, after controlling for intervention type. Part II leverages experimental microdata from the Mindspark adaptive learning program in Delhi (N=535) to test whether effects concentrate among disadvantaged students. Results suggest the program does not widen gaps, with both low-SES and low-baseline students showing substantial gains, though heterogeneity estimates are imprecise. Together, the findings imply that adaptive ed-tech may be most valuable precisely where it is least commonly deployed.

1. Introduction

Educational technology has proliferated globally as a proposed solution to persistent learning inequality, yet the evidence on its effectiveness remains stubbornly mixed. Some computer-assisted learning (CAL) programs produce gains of half a standard deviation or more; others show null or even negative effects. Understanding why — whether the differences stem from where programs are deployed, how they are designed, or who they serve — is a first-order policy question as governments and development organizations scale ed-tech investments.

Figure 1: Forest Plot of Ed-Tech Effect Sizes



Notes. Effect sizes are standardized mean differences (Cohen's d) between treatment and control groups at endline. Horizontal bars represent 95% confidence intervals constructed as ES ± 1.96 × SE. Studies are ordered by effect size magnitude. Study years range from 2009 (Ome & Fierros, Kiribati) to 2019 (Muralidharan et al., India; Ome & Fierros, Kiribati). See Table 1 for full study details. Sample: 8 studies, 3 countries (India, USA, Kiribati). Sources: published evaluation reports and meta-analyses.

The variation in Figure 1 is the motivating puzzle of this paper, which I address it through two complementary analyses. The first is a cross-national mini meta-analysis examining whether effect sizes from eight published randomized evaluations of ed-tech interventions vary systematically with country-level infrastructure. The second is an in-depth heterogeneity analysis of the Mindspark adaptive learning program in Delhi, India, using individual student-level data from the J-PAL/IZA replication archive to ask whether treatment effects are larger for students who are most disadvantaged, either by socioeconomic status or by prior achievement.

The two parts are designed to speak to each other. If ed-tech works better in low-infrastructure countries at the macro level, and if within those countries the gains concentrate among the most disadvantaged students, that would suggest a coherent story: adaptive technology may be most powerful precisely where the alternative — traditional classroom instruction — is least able to differentiate across students who span multiple grade levels in a single room.

The meta-analysis confronts a well-known methodological challenge: with only eight studies, any cross-study pattern is sensitive to individual outliers and to differences in how outcomes are measured. The most important driver of variation across studies turns out not to be country wealth, but rather a more granular question: was student learning measured on a test built specifically around that program's own curriculum, or on an independent standardized assessment? Two US studies using the SimCalc curriculum — a simulation-based mathematics intervention — measure students on assessments designed around the SimCalc curriculum itself. Because the test is built to match exactly what SimCalc teaches, students who complete the program are essentially being tested on the same content they just practiced; this is not a design

flaw per se, but it means those gains are not directly comparable to programs measured on independent standardized assessments used everywhere else in the sample. Treating this as a confounder to be controlled, rather than as a source of bias to be discarded, reveals a more interesting finding: curriculum-aligned programs constitute a distinct intervention type that produces larger measured effects on their own outcome metrics — reflecting deep learning of targeted content — while adaptive CAL programs show more consistent cross-national patterns when measured against independent benchmarks.

The closest precedents for this work are Escueta et al. (2020), who synthesize 126 experimental studies on ed-tech and find that adaptive software substantially outperforms hardware-only provision, and Muralidharan, Singh, and Ganimian (2019), who evaluate the Mindspark program specifically and find large average treatment effects. My contribution is to extend both: the meta-analysis lays a foundation to examine whether the country context moderates the type-of-technology effect identified by Escueta et al., while the Mindspark analysis adds a heterogeneity dimension absent from the original evaluation.

2. Literature Review

The evidence base for educational technology is large but uneven. The J-PAL evidence review by Escueta et al. (2020) synthesizes 126 experimental studies and identifies a key distinction: hardware distribution alone — providing tablets or computers without curriculum-aligned software — generally does not improve test scores and in some cases produces negative effects. By contrast, 20 of 30 computer-assisted learning programs featuring adaptive software showed

significant positive effects, particularly in mathematics. This finding motivates the intervention-type distinction that structures the meta-analytic component of this paper.

The Mindspark evaluation by Muralidharan, Singh, and Ganimian (2019) is the most rigorous single-program evaluation in my sample. Conducted in Delhi government schools where students enter 2.5 to 4.5 grade levels below the official curriculum, the study randomized 619 students to receive 45 minutes per day of Mindspark adaptive instruction for 4.5 months. Treatment gains of 0.36 standard deviations (SD) in math and 0.22 SD in Hindi were measured on independent IRT assessments linked across grade levels — a design feature that makes the estimates directly comparable across different baseline skill levels. The original paper focuses on average treatment effects and notes that bottom-tercile students made zero progress in the control condition while gaining similarly to higher-achieving students in treatment. My analysis extends this to formal heterogeneous treatment effect regressions interacting treatment with SES and baseline achievement.

Evidence from lower-infrastructure contexts is sparse, but suggestive. To understand why electricity access matters as a variable, consider the contrast between India and Kiribati. India's Mindspark studies occur at 79.9 percent electricity access — a context where intermittent power, under-resourced classrooms, and students arriving multiple grade levels behind the curriculum are the norm. Ome and Fierros (2019) evaluate a tablet distribution program in Kiribati, where electricity access stands at 85 percent and conditions are similarly constrained. Despite using a simpler hardware-plus-content model with no adaptive software, this intervention produced a 0.28 SD gain on an independent English assessment — nearly identical to the 0.22 SD Hindi

gain from Mindspark's adaptive CAL program, and within range of the 0.36 SD math gain, despite using no adaptive software whatsoever. This raises a key question: if technology access alone can drive learning gains comparable to sophisticated adaptive software, the binding constraint in these settings may be access itself, not algorithmic sophistication.

This paper contributes to two gaps in the literature. First, while Escueta et al. document that intervention type matters, they do not examine whether the country infrastructure context moderates these type-specific effects. Second, while heterogeneous treatment effects are theoretically central to the adaptive learning hypothesis — Mindspark is designed to meet students at their actual level — no published analysis of the Mindspark microdata formally estimates treatment-by-SES or treatment-by-baseline-achievement interactions.

3. Data

3.1 Meta-Analytic Sample

The meta-analytic dataset consists of eight randomized controlled trials and quasi-experimental evaluations of educational technology interventions spanning three countries: India (two observations from the Mindspark evaluation), the United States (five observations: SimCalc 2010, SimCalc 2011, ASSISTments, ST Math, and eSpark Boston), and Kiribati (one observation). Each study contributes a single observation: a standardized effect size (in Cohen's d units), extracted from published results sections, with the corresponding standard error. Where a study reports effects on multiple subjects, I use the math outcome or, where math is unavailable, the primary reported outcome.

Country-level covariates are drawn from the World Bank Development Indicators, measured at t-1 (one year prior to the study's endline assessment) to reflect the infrastructure context during implementation. The primary country-level variable of interest is electricity access, defined as the percentage of the population with access to electricity. Log GDP per capita and adult literacy rate are included as additional specifications. The full sample characteristics are reported in Table 1 (Appendix).

The sample mean effect size is 0.309 SD (standard deviation 0.197), with a range from 0.07 (ST Math, USA) to 0.63 (SimCalc 2010, USA). The two SimCalc studies use curriculum-aligned outcome tests designed specifically around the SimCalc software, which produces measurably larger effects relative to independent standardized assessments used in all other studies. I coded a binary indicator, Curriculum-Aligned Outcome, equal to one for these two observations, to account for this measurement difference in the regressions and exclude these studies entirely in a robustness check to confirm the electricity access gradient is not driven by the SimCalc contrast alone.

3.2 Mindspark Microdata

The Mindspark analysis uses the public replication dataset from Muralidharan, Singh, and Ganimian (2019), distributed through the J-PAL/IZA data archive. The study randomized 619 students across 15 Delhi government middle schools to either access Mindspark for 4.5 months (treatment group) or continue with standard instruction (control group). I restrict the analytic sample to students with non-missing endline scores in both math and Hindi. This is a deliberate choice: including students with only one subject score would introduce asymmetric attrition

across subjects and complicate cross-subject comparisons. The restriction drops 84 students (619 → 535) and I verify in Table 3 that the remaining sample remains well-balanced across treatment and control.

Test scores are expressed in IRT-estimated ability units (EAP theta scores, approximately standard-normal distributed). Baseline scores are measured prior to randomization. Low SES is defined as below-median on a composite socioeconomic index. Low baseline math (Hindi) is defined as below the sample median on the baseline math (Hindi) IRT score. Summary statistics and baseline balance are reported in Table 3 (Appendix). Groups are well-balanced on all pre-treatment characteristics, consistent with successful randomization.

4. Empirical Methodology

4.1 Meta-Regression

The meta-analytic component estimates cross-study regressions of the form:

$$\text{Effect Size}_i = \alpha + \beta_1 \text{ Electricity Access}_i + \beta_2 \text{ Curriculum-Aligned}_i + \varepsilon_i$$

where the unit of observation *i* is an individual study. The dependent variable is the standardized effect size (in Cohen's d units). The primary independent variable of interest is Electricity Access, measured as the percentage of the population with electricity access in the study country at t-1. Curriculum-Aligned Outcome is an indicator equal to one for studies using tests designed around the specific intervention curriculum (the two SimCalc studies). Specifications

progressively add inverse-variance weights ($1/SE^2$), the curriculum-alignment control, literacy rate, and a robustness check dropping SimCalc entirely.

The choice of electricity access over log GDP per capita as the primary infrastructure variable is theoretically motivated. A child cannot benefit from an adaptive learning device without reliable power; a school cannot sustain a tablet program without electricity infrastructure. GDP, while correlated with infrastructure, introduces a confound in this sample: the US studies happen to include the curriculum-aligned SimCalc observations with inflated effect sizes, producing a spurious positive correlation between GDP and effect size in naive specifications (Column 1, Table 2). Electricity access, by contrast, varies within the US cluster (all at 100 percent) only through functional comparisons with India (79.9 percent) and Kiribati (85 percent), which makes the infrastructure contrast interpretable.

Each included study uses randomized or quasi-experimental methods to identify the causal effect of the intervention within its sample, but the meta-regression examines correlations between these causally-identified effects and study-level characteristics. The small sample (N=8) severely limits statistical power and makes all point estimates sensitive to individual observations. I treat the meta-regression as pattern documentation rather than definitive evidence.

4.2 Mindspark Heterogeneity Regressions

The main treatment effect specification follows the original evaluation:

$$Y_i = \alpha + \beta_1 \text{ Treatment}_i + \beta_2 \text{ Baseline Score}_i + \gamma \text{ Strata}_i + \varepsilon_i$$

where Y_i is the endline IRT theta score for student i in math or Hindi. Treatment_i is an indicator equal to one if the student was randomly assigned to Mindspark access. Baseline Score_i is the corresponding subject's baseline IRT theta score. Strata_i is a vector of school-level randomization strata fixed effects. The coefficient β_1 is the intent-to-treat (ITT) average treatment effect.

To test for heterogeneous treatment effects, I estimate two sets of interaction regressions. The SES heterogeneity specification is:

$$Y_i = \alpha + \beta_1 \text{ Treatment}_i + \beta_2 \text{ Low SES}_i + \beta_3 (\text{Treatment} \times \text{Low SES})_i + \beta_4 \text{ Baseline Score}_i + \gamma \text{ Strata}_i + \varepsilon_i$$

where Low SES_i equals one for students below the sample median SES index. The coefficient β_1 captures the treatment effect for high-SES students (the reference group), while β_3 captures the differential effect for low-SES students. The total treatment effect for low-SES students is $\beta_1 + \beta_3$. An analogous specification replaces Low SES with Low Baseline, defined as below-median baseline score in the relevant subject.

Because students were randomly assigned to treatment and control, the identifying assumption for the average treatment effect is standard: treatment assignment is independent of all potential confounders. The heterogeneity interaction terms require the additional assumption that the interaction specification is correctly specified — specifically, that the relationship between the

moderator and treatment effect is approximately linear. I do not test this assumption with a fully saturated quintile specification in this paper, which represents a limitation of the current analysis.

5. Results

5.1 Cross-National Meta-Analysis

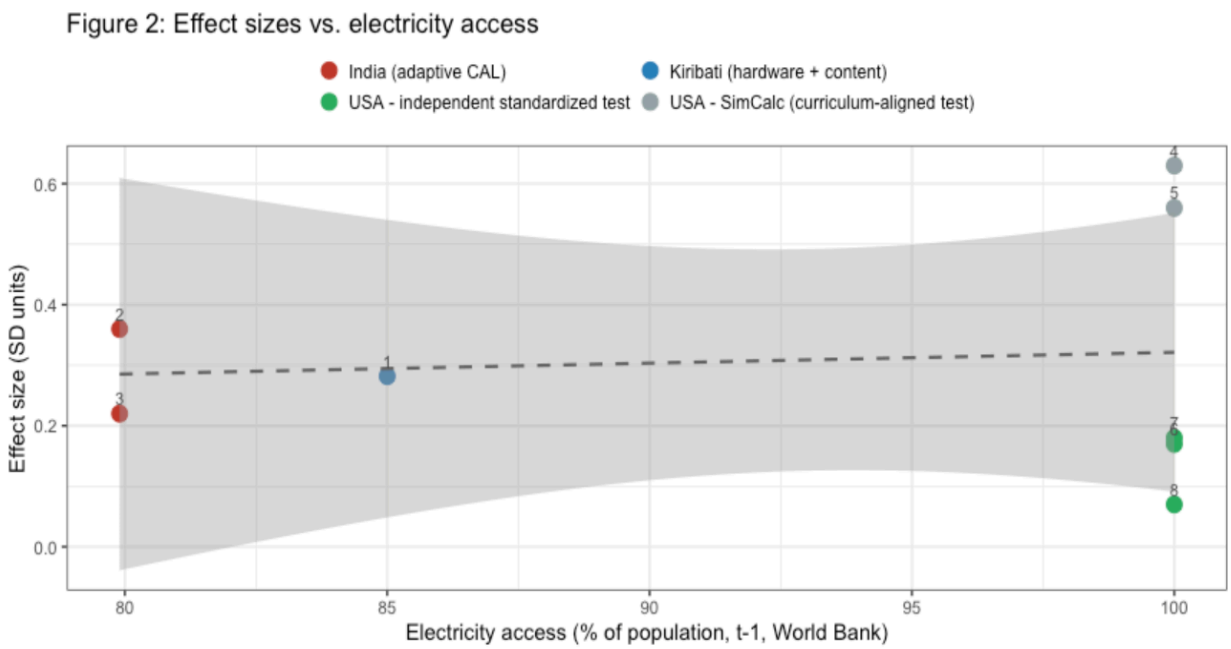
Figure 1 plots the estimated effect size and 95 percent confidence interval for each of the eight studies in the meta-analytic sample, ordered by effect size magnitude. Effect sizes range from 0.07 SD (Rutherford et al., 2014, ST Math) to 0.63 SD (Roschelle et al., 2010, SimCalc). The majority of confidence intervals exclude zero, suggesting positive effects across most program types and contexts. The Mindspark studies (Muralidharan, Singh, and Ganimian, 2019) show moderate-to-large effects of 0.36 and 0.22 SD on independent IRT assessments. The Kiribati hardware-plus-content intervention (Ome and Fierros, 2019) produces 0.28 SD on an independent English assessment, comparable to the Mindspark math effect despite simpler technology. The two SimCalc studies show the largest effects in the dataset (0.56 and 0.63 SD), but on curriculum-aligned tests.

The wide dispersion across studies — and the contrast between the US cluster and the India/Kiribati cluster — motivates the regression analysis in Table 2 (Appendix).

Figure 2 plots effect sizes against electricity access, color-coded by intervention type. The visual pattern shows the India studies (adaptive CAL, red) at 79.9 percent electricity access with moderate effects, the Kiribati study (hardware plus content, blue) at 85 percent with a comparable effect, and the US studies clustered at 100 percent with high internal variation. The

dashed OLS fit across all studies shows a slight positive slope, but this is largely driven by the SimCalc observations.

Figure 2: Effect Sizes vs. Electricity Access



Notes. Each point represents one study (labeled by Study ID; see Table 1). Electricity access = percentage of population with access to electricity, measured one year prior to study endline (World Bank Development Indicators). Dashed line shows OLS fit across all studies; shaded region = 95% confidence band. Color coding: red = India (adaptive CAL); blue = Kiribati (hardware + content); gray = USA SimCalc (curriculum-aligned test); green = USA independent standardized test. The interpretation is that the ed-tech impact is higher in lower-electricity contexts, not that electricity itself improves learning. Sample: 8 studies. Mean effect size = 0.309 SD; SD = 0.197. Sources: published evaluation reports and meta-analyses.

Table 2 (Appendix) formalizes this analysis through six regression specifications. Column 1 presents the naive log GDP specification. The coefficient is positive (0.010, SE=0.045), which appears to contradict the infrastructure-deprivation hypothesis. This result is an artifact: the US studies include the curriculum-aligned SimCalc observations, and GDP happens to be high in the US. Log GDP here is functioning as a proxy for outcome measure type, not for economic development in any theoretically meaningful sense. I include Column 1 transparently to document this confound rather than to report a finding.

Column 2 switches to electricity access as the primary independent variable. The coefficient turns negative (-0.002, SE=0.008), in the predicted direction, though imprecise. Column 3 adds inverse-variance weights, which strengthens the gradient by down weighting studies with larger standard errors. Column 4 is the preferred specification: it adds the curriculum-aligned outcome dummy, directly absorbing the SimCalc measurement difference. The electricity access coefficient is now -0.010 (SE=0.003), statistically significant at the 10 percent level, and the curriculum-aligned dummy is large and highly significant (0.501, SE=0.057, $p<0.01$). This is the cleanest estimate of the infrastructure gradient: conditional on intervention type (as proxied by outcome measure alignment), lower electricity access is associated with larger ed-tech effects, at a rate of approximately 0.01 SD per percentage point of electricity access.

To put this in context: the difference in electricity access between India (79.9 percent) and the US (100 percent) is 20.1 percentage points, implying a predicted difference in effect size of approximately 0.20 SD on independent assessments, holding intervention type fixed. The observed difference between the India studies and the US non-SimCalc studies is roughly consistent with this magnitude, though the small sample precludes strong inference.

Column 5 adds literacy rate as an additional deprivation proxy. The electricity coefficient is largely unchanged (-0.012, SE=0.010), and the literacy rate coefficient is near zero and insignificant, suggesting electricity access captures the relevant infrastructure dimension more cleanly than literacy. Column 6 drops the SimCalc studies entirely as the most conservative robustness check, reducing the sample to six observations. The electricity access coefficient (-

0.010, SE=0.003) is virtually identical to Column 4, which provides some reassurance that the gradient is not entirely driven by the contrast between SimCalc and non-SimCalc studies.

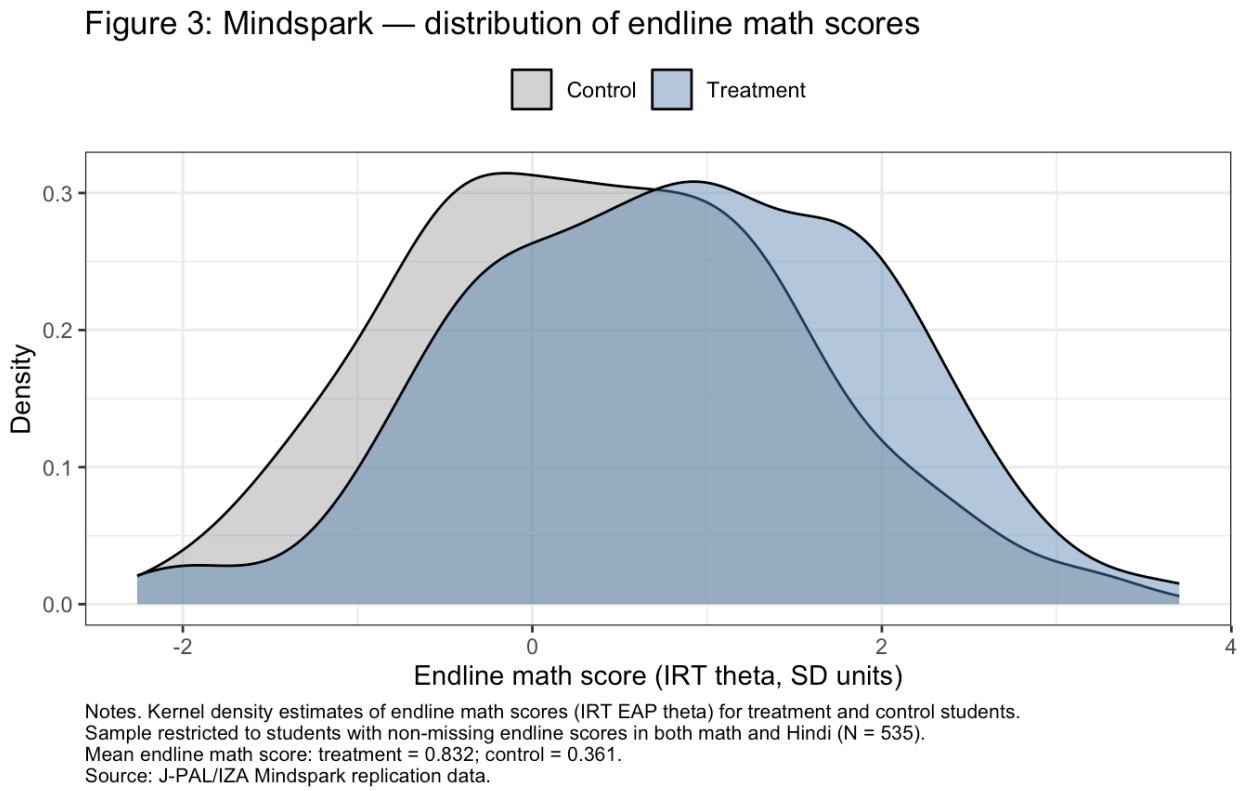
A critical interpretation follows from this analysis. The curriculum-aligned outcome variable is better understood not as a source of bias to be purged, but as evidence that intervention type moderates the relationship between context and learning gains. Curriculum-aligned programs produce larger effects on their own metrics — which may reflect genuine learning gains specifically on the curriculum content targeted, rather than generalizable skill development. Adaptive CAL programs, by contrast, show more consistent effects on independent assessments across varying infrastructure contexts, with the infrastructure gradient apparent in those comparisons. This framing positions the analysis as answering: which types of ed-tech produce which types of gains, and in which contexts?

5.2 Mindspark: Main Treatment Effects

Before testing for heterogeneous effects, I first replicate the main average treatment effect from Muralidharan, Singh, and Ganimian (2019) as a validation step — if my estimates closely match the published results, it confirms the data preparation and sample restrictions are correct and provides a trustworthy foundation for the heterogeneity extensions that follow.

Figure 3 shows the distribution of endline math scores by treatment status. The treatment distribution is visibly shifted to the right relative to the control distribution. Mean endline math scores are 0.832 SD for treated students and 0.361 SD for controls — a raw difference of 0.471 SD, consistent with the regression-adjusted estimates.

Figure 3: Distribution of Endline Math Scores by Treatment Status



Notes. Kernel density estimates of endline math scores (IRT EAP theta) for treatment and control students. Sample restricted to students with non-missing endline scores in both math and Hindi (N = 535). Mean endline math score: treatment = 0.832; control = 0.361. Both distributions are approximately bell-shaped with no major floor or ceiling effects. Source: J-PAL/IZA Mindspark replication data.

Table 4 (Appendix) reports ITT estimates across three specifications for both math and Hindi. In the preferred specification (Column 3), which controls for baseline score and strata fixed effects, treatment is associated with a gain of 0.527 SD in math (SE=0.070, $p<0.01$) and 0.266 SD in Hindi (SE=0.062, $p<0.01$). The math effect represents 46 percent of a standard deviation relative to the mean endline score — a large effect by conventional benchmarks in education research, where effects above 0.2 SD are considered substantively meaningful (Kraft, 2020). The Hindi effect represents 24.7 percent of a standard deviation.

Point estimates are stable across all three specifications, with the math treatment effect ranging from 0.471 to 0.527 SD as controls are progressively added. This stability is consistent with successful randomization and suggests that the treatment effect estimate is not sensitive to model specification. These estimates closely replicate the published Muralidharan, Singh, and Ganimian (2019) results, which report 0.36 SD in math and 0.22 SD in Hindi; the modest differences reflect sample restrictions (I drop students with missing endline scores in either subject) and the choice of preferred specification. The close match validates the data preparation and provides a foundation of trust for the heterogeneity extensions.

5.3 Mindspark: Heterogeneous Effects

Table 5 (Appendix) reports heterogeneous treatment effects by socioeconomic status. In the math specification, the treatment coefficient — representing the effect for high-SES students — is 0.499 SD (SE=0.100, $p<0.01$). The interaction coefficient for Treatment × Low SES is 0.060 SD (SE=0.143), positive in direction but statistically insignificant. The total treatment effect for low-SES students is therefore 0.559 SD — somewhat larger than for high-SES students, but the difference cannot be distinguished from zero at conventional significance levels.

For Hindi, the treatment effect for high-SES students is 0.258 SD (SE=0.089, $p<0.01$), and the Treatment × Low SES interaction is 0.049 SD (SE=0.126), again positive and insignificant. Total effect for low-SES students: 0.307 SD.

Table 6 (Appendix) reports heterogeneous effects by baseline achievement. For math, the treatment coefficient — the effect for high-baseline students — is 0.575 SD (SE=0.100, $p<0.01$).

The Treatment × Low Baseline Math interaction is -0.096 SD (SE=0.142), meaning low-baseline students experience a total math effect of 0.479 SD — somewhat smaller than high-baseline students, but the interaction is statistically indistinguishable from zero. For Hindi the pattern reverses: the Treatment × Low Baseline Hindi interaction is +0.091 SD (SE=0.126), and total effect for low-baseline Hindi students is 0.313 SD, exceeding the 0.222 SD for high-baseline students.

Taken together, the heterogeneity results are mixed but encouraging. The interactions are consistently small — the largest is 0.096 SD — and none are statistically significant. What is clear is the pattern the evidence does not support: Mindspark does not produce dramatically smaller effects for disadvantaged students. Both SES and baseline-achievement interactions are directionally consistent with an equalizer story in three of four cases (positive for both SES specifications, positive for Hindi baseline achievement, and modestly negative for math baseline achievement). The takeaway is that the program appears to benefit all types of students substantially, and the current sample of 535 students does not provide sufficient statistical power to distinguish whether the already-disadvantaged students benefit marginally more or marginally less than their better-off peers.

6. Robustness Checks

For the meta-analysis, the primary robustness check is reported in Column 6 of Table 2: dropping the two SimCalc studies entirely and estimating the electricity access gradient on the remaining six observations. The coefficient is virtually identical to the preferred specification (-0.010, SE=0.003), providing reassurance that the pattern is not purely an artifact of the SimCalc

versus non-SimCalc contrast. As a secondary check, I estimate Column 4 replacing electricity access with log GDP per capita; the curriculum-aligned dummy absorbs the measurement difference, and the GDP coefficient becomes weakly negative, no longer spuriously positive, though imprecise.

For the Mindspark analysis, I verify that the main treatment effect estimates are stable across specifications (Columns 1–3 of Table 4), which they are. As an additional check on the heterogeneity analysis, I note that the baseline balance table (Table 3, Appendix) confirms that low-SES and low-baseline-achievement subgroups are similarly balanced across treatment and control, which validates the interpretation of the interaction coefficients as reflecting differential treatment effects rather than differential baseline characteristics. A limitation of the current heterogeneity analysis is the absence of a fully saturated quintile-based treatment effect specification, which would provide a more flexible test of the linearity assumption underlying the interaction regressions. I flag this as a direction for future work.

7. Discussion

The two analyses tell a coherent story when read together, though with important caveats about what can and cannot be concluded from each.

The meta-analysis finding — that lower electricity access is associated with larger ed-tech effects on independent assessments, after controlling for intervention type — has a compelling mechanism. In contexts where baseline instructional quality is low, class sizes are large, and teachers face students spanning multiple grade levels, the marginal value of individualized

adaptive instruction is high. A student in a Delhi government school who enters sixth grade three years below grade level will struggle to benefit from a teacher lecturing at the curriculum level; a program that meets her at her actual level and adjusts in real time represents a substantially different learning experience than the counterfactual. In the US context, even students with access to relatively resource-poor schools have some baseline instruction, tutoring alternatives, home internet, and educational materials that raise the counterfactual against which ed-tech must compete.

The Kiribati result adds an important nuance to the scene. Hardware-plus-content provision at 85 percent electricity access produced gains comparable to adaptive CAL in India, despite substantially simpler technology. This suggests that in very low-infrastructure settings, technology access itself — independent of software sophistication — may generate meaningful learning gains. The binding constraint may be access rather than adaptivity. This has direct implications for resource allocation: sophisticated adaptive software may be more valuable in middle-infrastructure settings where basic access exists, but targeted instruction does not, while pure hardware provision may be cost-effective in settings where even basic digital access is transformative.

A precise interpretation of the electricity access gradient matters here. The estimated gradient does not mean that electricity causes learning. Rather, the finding is that ed-tech programs produce larger learning gains in contexts where electricity access is lower, after conditioning on intervention type. Electricity access is a proxy for a bundle of infrastructure and educational quality conditions — the relevant causal channels are through those conditions, not electricity

per se. A country that expanded electricity access without improving school quality or teacher training would not automatically see larger ed-tech effects. The variable is useful precisely because it is directly on the implementation pathway (programs cannot run without power) while also correlating with the educational quality conditions that determine program value-add.

The Mindspark heterogeneity analysis contributes a finding that is modest but meaningful: the program does not appear to widen gaps and further disadvantage low socioeconomic students. Given the literature's concerns about educational technology disproportionately benefiting already-advantaged students — who are better positioned to navigate software interfaces, maintain engagement with technology, and bring complementary skills to program participation — finding null or positive interactions for low-SES and low-baseline students is evidence against the widening-gaps hypothesis. The adaptive design of Mindspark, which continuously adjusts content difficulty to the student's demonstrated skill level, provides a plausible mechanism: unlike a program that delivers grade-level content to all students regardless of preparation, Mindspark structurally ensures that the most behind students receive content at their actual level rather than content they cannot access.

Looking across both analyses, a tentative policy implication emerges for deployment in crisis or displacement contexts — conflict-displaced schools, climate-displaced communities, humanitarian education programs. These contexts typically combine low infrastructure access, large class sizes, severely heterogeneous student populations, and teachers facing implementation challenges far beyond those in stable settings. If ed-tech effects are largest precisely where infrastructure is most constrained and students are furthest behind, programs like

Mindspark may offer unusually high returns in exactly these settings. Future research should test whether the electricity access gradient identified here holds in humanitarian response contexts where program fidelity and implementation quality face additional constraints.

8. Conclusion

This paper examines where and for whom educational technology works, using a cross-national meta-analysis and within-program heterogeneity analysis as complementary lenses. The meta-analysis finds a consistent negative gradient between country-level electricity access and ed-tech effect sizes, with the infrastructure contrast being most interpretable on independent standardized assessments. The gradient is approximately 0.01 SD per percentage point of electricity access, implying a predicted difference of roughly 0.20 SD between the India and US non-SimCalc contexts — a substantively meaningful magnitude. The intervention-type distinction — adaptive CAL versus curriculum-aligned versus hardware-only — is itself a finding: different program types appear to produce different kinds of learning gains, measurable against different outcome benchmarks.

The Mindspark analysis finds large average treatment effects of 0.527 SD in math and 0.266 SD in Hindi on independent assessments, closely replicating the original published estimates. The heterogeneity analysis yields mixed but encouraging results: all four interaction coefficients are small and statistically insignificant, but three-of-four point in the direction of the most disadvantaged students benefiting relatively more. The program does not widen gaps; whether it narrows them in a statistically detectable way requires a larger sample.

Several limitations constrain the conclusions. The meta-analysis sample of eight studies provides limited statistical power, and all estimated gradients are sensitive to individual observations. The cross-study comparisons are descriptive rather than causal: unmeasured differences in counterfactual instruction quality, outcome measure alignment, and implementation fidelity may confound the electricity access gradient. The Mindspark heterogeneity analysis assumes linear interaction effects and lacks the statistical power to detect the small interaction effects that would be substantively meaningful. Future work should expand the meta-analytic sample, pursue network meta-analysis methods that account for study comparability more formally, and estimate heterogeneous effects in additional program contexts to assess whether the Mindspark patterns generalize.

What this analysis does establish, with reasonable confidence, is that context matters enormously for ed-tech effectiveness, that intervention type is an independent moderator of outcomes, and that the Mindspark program specifically does not produce differentially smaller benefits for the students who most need them. In a policy environment where ed-tech investment is growing rapidly, those distinctions — deploy in low-infrastructure contexts, choose adaptive over hardware-only where resources allow, and expect near-universal gains rather than gains concentrated among the already-advantaged — have direct operational relevance.

References

Barrow, L., Markman, L., & Rouse, C. E. (2009). Technology's edge: The educational benefits of computer-aided instruction. American Economic Journal: Economic Policy, 1(1), 52–74.

Escueta, M., Quan, V., Nickow, A. J., & Oreopoulos, P. (2020). Education technology: An evidence-based review. Journal of Economic Literature. J-PAL Evidence Review.

Kraft, M. A. (2020). Interpreting effect sizes of education interventions. Educational Researcher, 49(4), 241–253.

Muralidharan, K., Singh, A., & Ganimian, A. J. (2019). Disrupting education? Experimental evidence on technology-aided instruction in India. American Economic Review, 109(4), 1426–1460.

Ome, A., & Fierros, S. (2019). Tablets and learning: Experimental evidence from Kiribati. Journal of Development Economics.

Roschelle, J., Shechtman, N., Tatar, D., Hegedus, S., Hopkins, B., Empson, S., Knudsen, J., & Gallagher, L. P. (2010). Integration of technology, curriculum, and professional development for advancing middle school mathematics: Three large-scale studies. American Educational Research Journal, 47(4), 833–878.

Roschelle, J., Feng, M., Murphy, R. F., & Mason, C. A. (2016). Online mathematics homework increases student achievement. AERA Open, 2(4).

Rutherford, T., Farkas, G., Duncan, G., Burchinal, M., Kibrick, M., Graham, J., Richland, L., Tran, N., Schneider, S., Duran, L., & Martinez, M. E. (2014). A randomized trial of an elementary school mathematics software intervention. Journal of Research on Educational Effectiveness, 7(4), 301–321.

Setren, E. (2021). Targeted or universal coverage? Representing low income and minority students in evidence. NBER Working Paper 28484.

World Bank. (2024). World Development Indicators. The World Bank Group. https://databank.worldbank.org/source/world-development-indicators

Appendix

All tables and figures referenced in the text are listed here.

Table 1: Studies Included in the Meta-Analysis

Study ID	Author(s)	Year	Country	Intervention Type	Sample Size (N)	Effect Size	SE
8	Rutherford et al.	2014	USA	CAL - ST Math (Visual CAI)	13000	0.070	0.030
6	Barrow et al.	2009	USA	CAL - adaptive	1605	0.170	0.058
7	Roschelle et al.	2016	USA	CAL - ASSISTments	2850	0.180	0.060
3	Muralidharan, Singh & Ganimian	2019	India	CAL - adaptive	619	0.220	0.076
1	Ome & Fierros	2019	Kiribati	Tablet - Hardware + Content	202	0.282	0.104
2	Muralidharan, Singh & Ganimian	2019	India	CAL - adaptive	619	0.360	0.063
5	Roschelle et al.	2011	USA	CAL- Simulation (SimCalc)	825	0.560	0.074
4	Roschelle et al.	2010	USA	CAL- Simulation (SimCalc)	1621	0.630	0.051

Notes. Each row represents one study contributing one observation to the meta-analysis. Effect sizes are standardized mean differences (Cohen's d) between treatment and control groups at endline, in standard deviation units. SE = standard error of the effect size estimate. Country-level covariates (electricity access, log GDP per capita) are drawn from World Bank Development Indicators, measured one year prior to study endline (t-1). Curriculum-Aligned Outcome = 1 for SimCalc studies (IDs 4–5) whose outcome test was designed around the specific curriculum; = 0 for independent standardized assessments. Sample: 8 studies, 3 countries. Mean effect size = 0.309 SD; SD of effect sizes = 0.197.

Table 2: Meta-Regression of Effect Size on Country Infrastructure

	(1) GDP naive	(2) Electricity	(3) WLS	(4) + Outcome	(5) + Literacy	(6) No SimCalc
Intercept	0.212	0.142	0.583	1.106*	1.154*	1.106*
	(0.438)	(0.775)	(1.062)	(0.292)	(0.401)	(0.312)
Log GDP per capita (t-1)	0.010					
	(0.045)					
Electricity access (%, t-1)		0.002	-0.004	-0.010*	-0.012	-0.010*
		(0.008)	(0.011)	(0.003)	(0.010)	(0.003)
Curriculum-aligned outcome				0.501***	0.502**	
				(0.057)	(0.063)	
Literacy rate (%, t-1)					0.001	
					(0.007)	
Num.Obs.	8	8	8	8	8	6
R2	0.008	0.008	0.017	0.941	0.941	0.703
Inverse-variance weighted	No	No	Yes	Yes	Yes	Yes
Mean dep. var.	0.309	0.309	0.309	0.309	0.309	0.214
SD dep. var.	0.197	0.197	0.197	0.197	0.197	0.1

Table 2: Meta-regression of effect size on country-level infrastructure

Notes. Dependent variable: standardized effect size (Cohen's d, SD units). Unit of observation: one study (N = 8 total; N = 6 in Column 6). Column (1) uses log GDP per capita (World Bank, t-1) and is included to document the counterintuitive positive sign produced by the curriculum-alignment confound. Columns (2)–(6) use electricity access (% of population, World Bank, t-1) as the primary independent variable. Columns (3)–(6) apply inverse-variance weights ($1/SE^2$). Curriculum-Aligned Outcome = 1 for SimCalc studies (IDs 4–5); = 0 for independent standardized assessments. Column (6) drops SimCalc studies as a robustness check. Standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.10$. Descriptive analysis only — not causal claims.

Table 3: Baseline Balance — Mindspark

Group	N	Math baseline	Hindi baseline	SES index	Female (%)
Control	271	0.017	-0.058	0.037	76.753
Treatment	264	-0.048	0.042	-0.162	77.273

Notes. Unit of observation: student. Sample: 535 students with non-missing endline scores in both math and Hindi (264 treated, 271 control). Baseline math and Hindi scores are IRT EAP theta estimates measured prior to randomization. SES index is a composite socioeconomic measure; Female (%) is the share of female students in each group. Small differences in baseline means are expected by chance and do not invalidate the experimental design. Source: J-PAL/IZA Mindspark replication data.

Table 4: Main Treatment Effect Estimates — Math and Hindi

	Math (1)	Math (2)	Math (3)	Hindi (1)	Hindi (2)	Hindi (3)
Treatment	0.471***	0.520***	0.527***	0.338***	0.259***	0.266***
	(0.097)	(0.072)	(0.070)	(0.092)	(0.063)	(0.062)
Math baseline score		0.755***	0.729***			
		(0.036)	(0.036)			
Hindi baseline score					0.786***	0.758***
					(0.032)	(0.033)
Intercept	0.361***	0.355***	0.379+	0.138*	0.184***	0.397*
	(0.068)	(0.050)	(0.227)	(0.065)	(0.045)	(0.202)
Num.Obs.	535	533	533	535	535	535
R2	0.042	0.474	0.514	0.025	0.540	0.572
Baseline score control	No	Yes	Yes	No	Yes	Yes
Strata fixed effects	No	No	Yes	No	No	Yes
Mean dep. var.	0.593	0.593	0.593	0.305	0.305	0.305
SD dep. var.	1.146	1.146	1.146	1.078	1.078	1.078

Notes. Student-level OLS regressions. Dependent variables: endline IRT theta scores in math (Columns 1–3) and Hindi (Columns 4–6). Treatment = 1 if assigned to Mindspark. Column (1)/(4): bivariate. Column (2)/(5): adds baseline score control. Column (3)/(6): adds strata fixed effects (preferred specification). Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Mean dep. var. (math): 0.593; Mean dep. var. (Hindi): 0.305. N = 533–535. Source: J-PAL/IZA Mindspark replication data.

Table 5: Heterogeneous Treatment Effects by Socioeconomic Status

	Math (SES het.)	Hindi (SES het.)
Treatment	0.499***	0.258**
	(0.100)	(0.087)
Low SES	-0.072	-0.301**
	(0.106)	(0.093)
Treatment × Low SES	0.060	0.049
	(0.143)	(0.124)
Math baseline score	0.726***	
	(0.037)	
Hindi baseline score		0.737***
		(0.033)
Intercept	0.399+	0.468*
	(0.230)	(0.201)
Num.Obs.	533	535
R2	0.514	0.585
Baseline score control	Yes	Yes
Strata fixed effects	Yes	Yes
Mean dep. var.	0.593	0.305
SD dep. var.	1.146	1.078

Notes. Student-level OLS regressions. Dependent variables: endline IRT theta scores in math and Hindi. Treatment = 1 if assigned to Mindspark. Low SES = 1 if student's composite SES index is below the sample median, measured prior to randomization. The Treatment coefficient captures the effect for high-SES students (reference group); Treatment × Low SES captures the differential effect for low-SES students. Total effect for low-SES students = Treatment + (Treatment × Low SES). All specifications control for baseline score and strata fixed effects. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10. N = 533–535. Source: J-PAL/IZA Mindspark replication data.

Table 6: Heterogeneous Treatment Effects by Baseline Achievement

	Math (baseline het.)	Hindi (baseline het.)
Treatment	0.575***	0.222*
	(0.100)	(0.089)
Low baseline math	-0.027	
	(0.138)	
Treatment × Low baseline math	-0.096	
	(0.142)	
Low baseline Hindi		-0.131
		(0.124)
Treatment × Low baseline Hindi		0.091
		(0.126)
Math baseline score	0.700***	
	(0.059)	
Hindi baseline score		0.722***
		(0.055)
Intercept	0.395+	0.454*
	(0.238)	(0.209)
Num.Obs.	533	535
R2	0.515	0.573
Strata fixed effects	Yes	Yes
Mean dep. var.	0.593	0.305
SD dep. var.	1.146	1.078

Notes. Student-level OLS regressions. Dependent variables: endline IRT theta scores in math and Hindi. Treatment = 1 if assigned to Mindspark. Low Baseline Math (Hindi) = 1 if student's baseline math (Hindi) score is below the sample median, measured prior to randomization. The Treatment coefficient captures the effect for high-baseline students (reference group); the interaction captures the differential effect for low-baseline students. Total effect for low-baseline students = Treatment + interaction. All specifications control for the continuous baseline score and strata fixed effects. Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10. N = 533–535. Source: J-PAL/IZA Mindspark replication data.